

7 SEPTEMBER 2026 - 31 MARCH 2027

Weekly Study Guide

A day-by-day plan synchronized with the technical syllabus and the four-project portfolio

DESIGN PROMISE

Every week names the syllabus module, the active project milestone, the required daily work and the evidence to save.

17 hours per full week

Monday to Sunday

25 core weeks plus March consolidation

Exact technical and project outputs

Prepared for Emmanuel Januarie

Study period: 7 September 2026 to 31 March 2027

Workload: 17 hours per full week

How the three PDFs work together

Use this guide as the calendar. Read the named module in the Technical Syllabus, then build the named milestone in Four Golden Portfolio Projects. Do not replace technical study with portfolio polishing.

DOCUMENT	QUESTION IT ANSWERS
Technical Syllabus	What must I understand, practise and troubleshoot?
Weekly Study Guide	What do I do today, for how long, and what do I save?
Portfolio Projects	How do I combine the skills into four strong, explainable builds?

Fixed weekly rhythm

Day	Time	Structure
Monday	2 h	20 recall 60 learn 40 calculate/draw
Tuesday	2 h	20 recall 50 learn 50 program/test
Wednesday	2 h	20 recall 60 learn 40 apply
Thursday	2 h	20 recall 60 learn 40 process case
Friday	2 h	20 review 50 rotate 40 practical 10 file
Saturday	4 h	60 plan 120 build 60 test/document
Sunday	3 h	60 closed-book recall 60 fault 60 evidence review

MINIMUM VIABLE WEEK

If life interrupts, protect Tuesday PLC practice, Saturday build/test work and Sunday recall/fault review. Reschedule foundations; do not quietly skip them.

Daily completion standard

Monday - Electrical

Recall last mistakes; learn one tight topic; solve at least five calculations or trace one circuit; record units and safe test steps.

Tuesday - PLC

Explain the last routine from memory; write or modify logic; test start, stop, fault and restart; save commented evidence.

Wednesday - Instrumentation

Trace sensor to HMI; calculate, draw or compare; state likely faults and safe checks.

Thursday - Process control

Link the concept to a conveyor, tank or water process; write one operator action and one automatic response.

Friday - Rotation

Complete only the dated A/B/C/D topic; save one artifact for Saturday integration.

Saturday - Project

Write acceptance criteria; build; integrate; test expected versus actual; record next smallest action.

Sunday - Review

Closed-book recall; one systematic fault; update wrong-answer log, test record and evidence index.

EVIDENCE LABELS

Studied | Simulated | Implemented | Tested | Documented | Certified. Use a label only when its evidence standard is actually met.

Friday rotation

A = HMI/SCADA; B = Industrial networking; C = Technical documentation; D = Integration and review. The cycle begins Friday, 11 September 2026 and repeats continuously. 25 December and 1 January are deliberately light.

Week 01 | 07 Sep - 13 Sep 2026

Electrical fundamentals

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M01	None - foundation	Foundation evidence pack

Day / time	Technical focus	Required result
Mon - 2 h 07 Sep	Quantities and safety	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 08 Sep	PLC purpose preview	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 09 Sep	Signal-chain preview	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 10 Sep	PV/SP preview	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 11 Sep	A - HMI/SCADA: HMI purpose and screen hierarchy	Create one rotation artifact and link it to the active build.
Sat - 4 h 12 Sep	Build low-voltage circuit evidence	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 13 Sep	Panel-indicator fault and filing	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Foundation evidence pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 02 | 14 Sep - 20 Sep 2026

Control circuits

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M02	None - foundation	Motor-control simulation

Day / time	Technical focus	Required result
Mon - 2 h 14 Sep	Relays, contactors and protection	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 15 Sep	Map relay logic to PLC	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 16 Sep	Switch states and feedback	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 17 Sep	Permissive chain	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 18 Sep	B - Networking: Ethernet and device addressing	Create one rotation artifact and link it to the active build.
Sat - 4 h 19 Sep	Simulate DOL start/stop circuit	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 20 Sep	Seal-in fault report	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Motor-control simulation ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 03 | 21 Sep - 27 Sep 2026

PLC hardware and I/O

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M03	Project 1 - Design	P1 design pack

Day / time	Technical focus	Required result
Mon - 2 h 21 Sep	24 VDC I/O paths	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 22 Sep	CPU, modules and scan	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 23 Sep	Digital input/output devices	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 24 Sep	Command and feedback path	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 25 Sep	C - Documentation: I/O list and tag convention	Create one rotation artifact and link it to the active build.
Sat - 4 h 26 Sep	Design conveyor I/O and architecture	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 27 Sep	Missing-input diagnosis	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 design pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 04 | 28 Sep - 04 Oct 2026

PLC addressing and diagnostics

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M03	Project 1 - Design	P1 signal trace

Day / time	Technical focus	Required result
Mon - 2 h 28 Sep	PNP/NPN awareness	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 29 Sep	Tags, monitor and backups	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 30 Sep	Analog I/O overview	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 01 Oct	Timing and short pulses	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 02 Oct	D - Integration/review: Integration review	Create one rotation artifact and link it to the active build.
Sat - 4 h 03 Oct	Complete signal tracing	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 04 Oct	Missing-output diagnosis	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 signal trace ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 05 | 05 Oct - 11 Oct 2026

Ladder Logic patterns

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M04	Project 1 - Logic	P1 logic library

Day / time	Technical focus	Required result
Mon - 2 h 05 Oct	Control-circuit comparison	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 06 Oct	Seal-in, timers and counters	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 07 Oct	Sensor edge behavior	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 08 Oct	Timing and state behavior	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 09 Oct	A - HMI/SCADA: Conveyor HMI controls	Create one rotation artifact and link it to the active build.
Sat - 4 h 10 Oct	Build logic library	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 11 Oct	Counter case and retest	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 logic library ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 06 | 12 Oct - 18 Oct 2026

Modes and permissives

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M05	Project 1 - Integrate	P1 modes

Day / time	Technical focus	Required result
Mon - 2 h 12 Oct	Motor-control review	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 13 Oct	Permissives and interlocks	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 14 Oct	Object detection	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 15 Oct	Modes and ownership	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 16 Oct	B - Networking: PLC/HMI communications path	Create one rotation artifact and link it to the active build.
Sat - 4 h 17 Oct	Add modes and start conditions	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 18 Oct	Will-not-start fault	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 modes ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 07 | 19 Oct - 25 Oct 2026

Sequences and feedback

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M05	Project 1 - Integrate	P1 integrated build

Day / time	Technical focus	Required result
Mon - 2 h 19 Oct	Overload protection	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 20 Oct	Sequence and counting	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 21 Oct	Sensor plausibility	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 22 Oct	Command-feedback timers	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 23 Oct	C - Documentation: Conveyor narrative and test plan	Create one rotation artifact and link it to the active build.
Sat - 4 h 24 Oct	Add sequence, faults and HMI	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 25 Oct	Sequence-stall case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 integrated build ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 08 | 26 Oct - 01 Nov 2026

Recovery and FAT

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M05	Project 1 - Prove	P1 final release

Day / time	Technical focus	Required result
Mon - 2 h 26 Oct	Isolation review	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 27 Oct	Refactor and comments	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 28 Oct	Signal plausibility	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 29 Oct	Safe recovery	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 30 Oct	D - Integration/review: Conveyor integration review	Create one rotation artifact and link it to the active build.
Sat - 4 h 31 Oct	Execute conveyor FAT	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 01 Nov	Retest and close defects	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P1 final release ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 09 | 02 Nov - 08 Nov 2026

Motors and starters

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M06	Project 2 - Actuation	P2 actuation pack

Day / time	Technical focus	Required result
Mon - 2 h 02 Nov	Motors and DOL starters	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 03 Nov	Motor interface logic	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 04 Nov	Running feedback	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 05 Nov	Command-feedback discrepancy	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 06 Nov	A - HMI/SCADA: Motor/VFD faceplate	Create one rotation artifact and link it to the active build.
Sat - 4 h 07 Nov	Build starter simulation	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 08 Nov	Overload diagnosis	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 actuation pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 10 | 09 Nov - 15 Nov 2026

VFD fundamentals

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M06	Project 2 - Actuation	P2 VFD interface

Day / time	Technical focus	Required result
Mon - 2 h 09 Nov	Nameplate and protection	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 10 Nov	Speed and reset logic	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 11 Nov	Analog speed reference	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 12 Nov	Drive status and trips	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 13 Nov	B - Networking: Drive communications awareness	Create one rotation artifact and link it to the active build.
Sat - 4 h 14 Nov	Complete VFD interface	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 15 Nov	LOCAL/Ready case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 VFD interface ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 11 | 16 Nov - 22 Nov 2026

Temperature and pressure

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M07	Project 2 - Measurement	P2 instrument list

Day / time	Technical focus	Required result
Mon - 2 h 16 Nov	Sensor power and wiring	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 17 Nov	Input diagnostics	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 18 Nov	Temperature/pressure selection	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 19 Nov	Measurement quality	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 20 Nov	C - Documentation: Instrument list and signal chains	Create one rotation artifact and link it to the active build.
Sat - 4 h 21 Nov	Select and document instruments	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 22 Nov	Frozen-signal drill	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 instrument list ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 12 | 23 Nov - 29 Nov 2026

Level, flow and process effects

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M07	Project 2 - Measurement	P2 selection matrix

Day / time	Technical focus	Required result
Mon - 2 h 23 Nov	Shielding awareness	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 24 Nov	Alarm from measurement	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 25 Nov	Level/flow technologies	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 26 Nov	Installation and process effects	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 27 Nov	D - Integration/review: Motor/sensor integration review	Create one rotation artifact and link it to the active build.
Sat - 4 h 28 Nov	Complete measurement chain	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 29 Nov	Ultrasonic-level case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 selection matrix ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 13 | 30 Nov - 06 Dec 2026

4-20 mA and scaling

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M08	Project 2 - Analog	P2 scaling proof

Day / time	Technical focus	Required result
Mon - 2 h 30 Nov	Analog wiring concepts	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 01 Dec	Raw values and scaling	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 02 Dec	Loop calculation	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 03 Dec	PV quality	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 04 Dec	A - HMI/SCADA: Analog faceplate, alarms and trends	Create one rotation artifact and link it to the active build.
Sat - 4 h 05 Dec	Build first analog loop	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 06 Dec	Open-loop diagnosis	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 scaling proof ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 14 | 07 Dec - 13 Dec 2026

Analog faults and records

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M08	Project 2 - Analog	P2 analog verification

Day / time	Technical focus	Required result
Mon - 2 h 07 Dec	Noise and shielding	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 08 Dec	Plausibility alarms	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 09 Dec	Calibration/HART awareness	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 10 Dec	Filtering and validation	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 11 Dec	B - Networking: Serial and HART fundamentals	Create one rotation artifact and link it to the active build.
Sat - 4 h 12 Dec	Test second loop	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 13 Dec	Wrong-scaling case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 analog verification ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 15 | 14 Dec - 20 Dec 2026

Valves and feedback loops

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M09	Project 2 - Process	P2 process build

Day / time	Technical focus	Required result
Mon - 2 h 14 Dec	Actuator interfaces	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 15 Dec	Tank sequence and modes	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 16 Dec	Valves and position feedback	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 17 Dec	PV/SP/MV and feedback	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 18 Dec	C - Documentation: Tank P&ID and alarm list	Create one rotation artifact and link it to the active build.
Sat - 4 h 19 Dec	Build tank process	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 20 Dec	Valve-will-not-open	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 process build ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 16 | 21 Dec - 27 Dec 2026

Alarms, trips and PID behavior

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M09	Project 2 - Process	P2 alarm/FAT prep

Day / time	Technical focus	Required result
Mon - 2 h 21 Dec	Protection review	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 22 Dec	Alarm/trip logic	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 23 Dec	Instrument faults	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 24 Dec	PID trends	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 25 Dec	D - Light catch-up / rest: Light catch-up and integration	Create one rotation artifact and link it to the active build.
Sat - 4 h 26 Dec	Add alarms and control	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 27 Dec	High-High case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 alarm/FAT prep ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 17 | 28 Dec - 03 Jan 2027

Process recovery and release

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M09	Project 2 - Prove	P2 final release

Day / time	Technical focus	Required result
Mon - 2 h 28 Dec	Schematic review	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 29 Dec	Refactor and comments	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 30 Dec	Range/plausibility audit	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 31 Dec	Manual/Auto recovery	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 01 Jan	A - Light HMI review / rest: Light HMI review	Create one rotation artifact and link it to the active build.
Sat - 4 h 02 Jan	Execute batch FAT	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 03 Jan	Close documents and release	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P2 final release ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 18 | 04 Jan - 10 Jan 2027

HMI and SCADA usability

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M10	Project 3 - HMI	P3 HMI pack

Day / time	Technical focus	Required result
Mon - 2 h 04 Jan	Panel/HMI power	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 05 Jan	HMI tag interface	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 06 Jan	PV quality display	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 07 Jan	Operator controls	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 08 Jan	B - Networking: IP plan and PLC/HMI path	Create one rotation artifact and link it to the active build.
Sat - 4 h 09 Jan	Build screen hierarchy	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 10 Jan	Stale-data case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P3 HMI pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 19 | 11 Jan - 17 Jan 2027

Ethernet and addressing

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M11	Project 3 - Network	P3 network baseline

Day / time	Technical focus	Required result
Mon - 2 h 11 Jan	Cables and topology	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 12 Jan	PLC Ethernet settings	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 13 Jan	Smart-instrument path	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 14 Jan	Data quality	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 15 Jan	C - Documentation: Network and protocol map	Create one rotation artifact and link it to the active build.
Sat - 4 h 16 Jan	Build IP plan and links	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 17 Jan	Link/IP fault	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P3 network baseline ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 20 | 18 Jan - 24 Jan 2027

Protocols and mapping

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M11	Project 3 - Network	P3 protocol proof

Day / time	Technical focus	Required result
Mon - 2 h 18 Jan	Shielding/termination awareness	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 19 Jan	Modbus and OPC mapping	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 20 Jan	Serial/HART review	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 21 Jan	Timeout and fallback	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 22 Jan	D - Integration/review: Communications recovery tests	Create one rotation artifact and link it to the active build.
Sat - 4 h 23 Jan	Execute protocol tests	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 24 Jan	Wrong-register case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P3 protocol proof ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 21 | 25 Jan - 31 Jan 2027

End-to-end SCADA integration

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M12	Project 3 - SCADA	P3 SCADA release

Day / time	Technical focus	Required result
Mon - 2 h 25 Jan	Control-panel review	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 26 Jan	PLC tag cleanup	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 27 Jan	Instrument quality	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 28 Jan	Integrated process path	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 29 Jan	A - HMI/SCADA: SCADA overview and diagnostics	Create one rotation artifact and link it to the active build.
Sat - 4 h 30 Jan	Integrate 20 tags and FAT	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 31 Jan	Read-only command case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P3 SCADA release ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 22 | 01 Feb - 07 Feb 2027

Troubleshooting and change control

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M13	Project 3 - Prove	P3 fault pack

Day / time	Technical focus	Required result
Mon - 2 h 01 Feb	Electrical test sequence	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 02 Feb	I/O and logic tracing	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 03 Feb	Loop isolation	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 04 Feb	Process symptoms	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 05 Feb	B - Networking: Layered network fault finding	Create one rotation artifact and link it to the active build.
Sat - 4 h 06 Feb	Run ten-fault pack	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 07 Feb	Pump selector case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P3 fault pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 23 | 08 Feb - 14 Feb 2027

Requirements and interfaces

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M14	Project 4 - Baseline	P4 baseline

Day / time	Technical focus	Required result
Mon - 2 h 08 Feb	Motor/valve interfaces	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 09 Feb	Equipment modules	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 10 Feb	Instrument definitions	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 11 Feb	Control narrative	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 12 Feb	C - Documentation: Flagship baseline documents	Create one rotation artifact and link it to the active build.
Sat - 4 h 13 Feb	Design first subsystem	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 14 Feb	Design review	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P4 baseline ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 24 | 15 Feb - 21 Feb 2027

Integration and failure behavior

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M14	Project 4 - Integrate	P4 integrated build

Day / time	Technical focus	Required result
Mon - 2 h 15 Feb	Protection audit	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 16 Feb	Sequences and modes	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 17 Feb	PV validation	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 18 Feb	Alarms and PID	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 19 Feb	D - Integration/review: Subsystem integration	Create one rotation artifact and link it to the active build.
Sat - 4 h 20 Feb	Integrate PLC/HMI/SCADA	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 21 Feb	Network-failure case	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P4 integrated build ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 25 | 22 Feb - 28 Feb 2027

Commissioning and handover

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
M14	Project 4 - FAT	P4 FAT release

Day / time	Technical focus	Required result
Mon - 2 h 22 Feb	Schematic final	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 23 Feb	Logic cleanup and backup	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 24 Feb	Tag audit	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 25 Feb	Recovery behavior	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 26 Feb	A - HMI/SCADA: Final HMI and trends	Create one rotation artifact and link it to the active build.
Sat - 4 h 27 Feb	Execute FAT and close defects	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 28 Feb	Technical sign-off	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P4 FAT release ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 26 | 01 Mar - 07 Mar 2027

Electrical and PLC consolidation

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
Review	Project 4 - Harden	P4 hardened build

Day / time	Technical focus	Required result
Mon - 2 h 01 Mar	Timed schematic trace	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 02 Mar	Timed logic-from-spec	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 03 Mar	Scaling drill	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 04 Mar	Cause-and-effect drill	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 05 Mar	B - Networking: Network troubleshooting explanation	Create one rotation artifact and link it to the active build.
Sat - 4 h 06 Mar	Fix highest-risk P4 defects	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 07 Mar	Regression tests	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ P4 hardened build ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 27 | 08 Mar - 14 Mar 2027

Instrumentation and SCADA consolidation

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
Review	Project 4 - Harden	Traceability pack

Day / time	Technical focus	Required result
Mon - 2 h 08 Mar	Control-path recall	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 09 Mar	PLC refactor	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 10 Mar	Instrument fault drill	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 11 Mar	Trend diagnosis	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 12 Mar	C - Documentation: Evidence and document audit	Create one rotation artifact and link it to the active build.
Sat - 4 h 13 Mar	Requirements-to-test audit	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 14 Mar	Fault-report rewrite	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Traceability pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 28 | 15 Mar - 21 Mar 2027

Integrated troubleshooting

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
Review	Project 4 - Present	Demo rehearsal 1

Day / time	Technical focus	Required result
Mon - 2 h 15 Mar	Electrical fault drill	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 16 Mar	PLC fault drill	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 17 Mar	Analog fault drill	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 18 Mar	Process fault drill	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 19 Mar	D - Integration/review: Mock technical demonstration	Create one rotation artifact and link it to the active build.
Sat - 4 h 20 Mar	Run integrated fault set	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 21 Mar	Correct weak explanations	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Demo rehearsal 1 ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 29 | 22 Mar - 28 Mar 2027

Targeted retest and release

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
Review	Project 4 - Present	Demo rehearsal 2

Day / time	Technical focus	Required result
Mon - 2 h 22 Mar	Weak electrical topic	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 23 Mar	Weak PLC topic	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 24 Mar	Weak instrument topic	Apply through a loop sketch, range calculation, signal trace or fault symptom.
Thu - 2 h 25 Mar	Weak process topic	Apply to the active process; state expected automatic and operator responses.
Fri - 2 h 26 Mar	A - HMI/SCADA: Final SCADA usability review	Create one rotation artifact and link it to the active build.
Sat - 4 h 27 Mar	Full regression and backup	Plan 60; build/integrate 120; test and document 60. Record expected vs actual.
Sun - 3 h 28 Mar	Final evidence audit	Recall 60; troubleshoot 60; file evidence and plan 60.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Demo rehearsal 2 ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Week 30 | 29 Mar - 31 Mar 2027

Final three-day close-out

SYLLABUS	PROJECT TRACK	WEEKLY OUTPUT
Close	Project 4 - Release	Final release pack

Day / time	Technical focus	Required result
Mon - 2 h 29 Mar	Final electrical recall	Recall, learn, then calculate, draw or trace. Save one corrected artifact.
Tue - 2 h 30 Mar	Final PLC and project demo	Write or inspect PLC logic line by line; test normal, stop, fault and restart.
Wed - 2 h 31 Mar	Final instrumentation recall	Apply through a loop sketch, range calculation, signal trace or fault symptom.

EVIDENCE TO FILE	WEEKLY SIGN-OFF
☐ Final release pack ☐ One PLC or diagnostic artifact ☐ One calculation/diagram/signal analysis ☐ One fault or case-study record	☐ Hours planned/completed recorded ☐ Correct module studied ☐ Correct project milestone advanced ☐ Normal and fault behavior tested ☐ Wrong-answer log updated ☐ Next smallest action written

Weekly record - print or duplicate

Week number / dates	_____
Syllabus module	_____
Active project milestone	_____
Hours planned / completed	_____
Main concept learned	_____
Practical output	_____
Fault investigated	_____
Evidence saved	_____
Weak point to review	_____
Next action	_____

DAILY SIGN-OFF ☐ Monday complete ☐ Tuesday complete ☐ Wednesday complete ☐ Thursday complete ☐ Friday rotation complete ☐ Saturday project complete ☐ Sunday review complete	QUALITY SIGN-OFF ☐ Expected/actual recorded ☐ Fault evidence saved ☐ Simulation labelled ☐ NOT TESTED used honestly ☐ Files versioned/backed up ☐ Next week planned
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RECOVERY RULE

If a week is missed, resume the next unfinished technical milestone. Do not skip prerequisite modules or double the next week's load. Reduce project scope before reducing safety, testing or documentation.